

Bull's AI

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Executive summary

Strategy & Innovation

A multi-strategy **Long-Short Equity** approach combining a sophisticated **Meta-Model Architecture** with a **risk-hedging Pairs Trading overlay**. The core innovation is leveraging **complementary signals** (market regime classification, quantitative factors, and textual sentiment) to maximize predictive power and portfolio robustness. (***Annualized Return: 26.34%*** (vs S&P 500: 12.95% to 15.29%))

Predictive Architecture

A **Meta-XGBoost** model integrates three diverse alpha sources: **Hidden Markov Model (HMM)** for market regime context; a second **XGBoost** for stock return prediction from **147 quantitative factors**; and **FinBERT** for sentiment from **SEC 10-K/Q filings**. (***Annualized Alpha: 38.30%***)

Key Performance

The strategy delivered on par performance with S&P 500 metrics over most of the range, except in 2015 and 2025, where there were significant excess returns and an extremely high Alpha over those periods. (***Information Ratio: 0.65***)

Key difference between Portfolio and S&P 500

Our portfolio's much higher annual standard deviation points to a very high volatility strategy (0.5840 vs 0.1528), resulting in a lower Sharpe ratio (0.5619 vs 0.7665), pointing to potential future work on incorporating quantitative risk metrics.

Investment Strategy

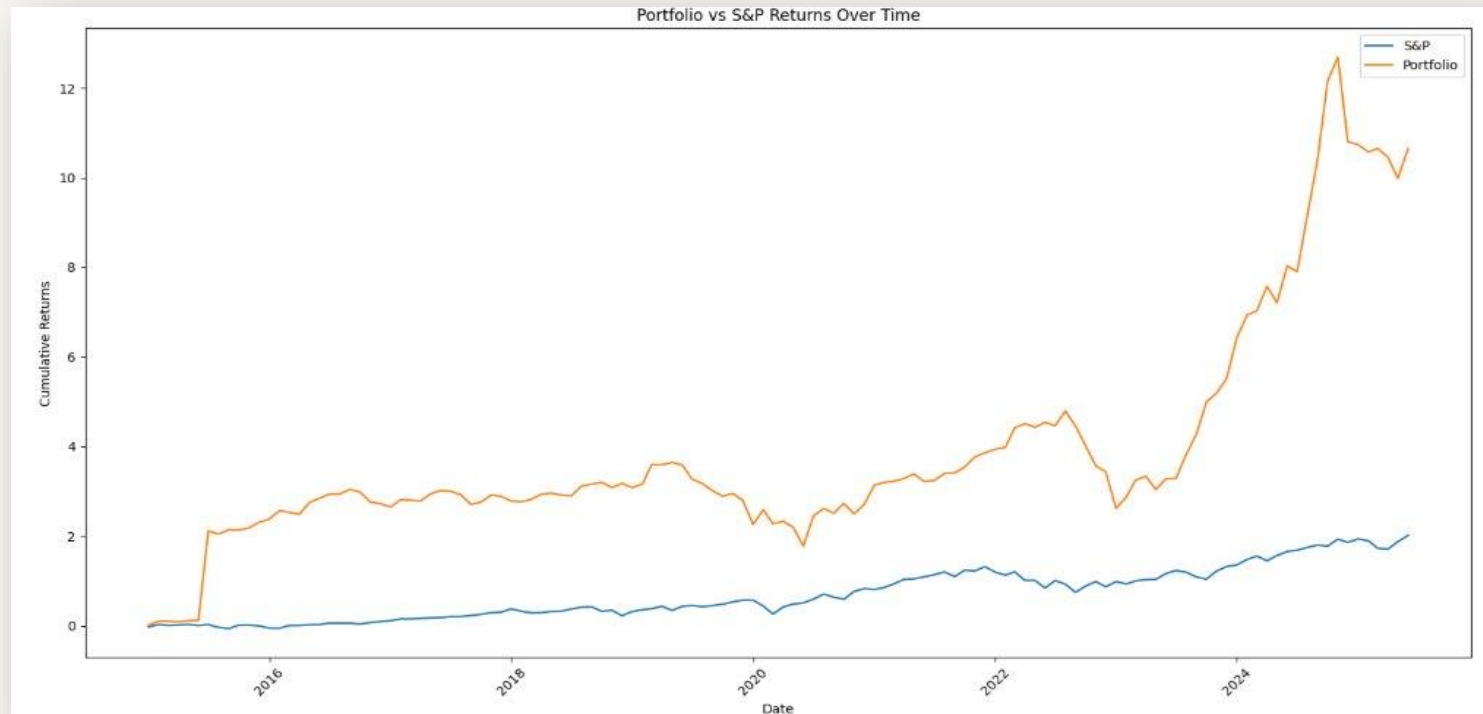
Our strategy combines multiple predictive models within a long-short framework. The core of the system is a meta-model (XGBoost) that integrates signals from:

- **Hidden Markov Model (HMM):** Market regime classification
- **XGBoost:** Stock return prediction
- **FinBERT (Transformer):** Sentiment analysis based on SEC Earnings reports (For US-based companies only)

The meta-model ranks all stocks based on expected returns. We go **long the top 100** and **short the bottom 100** stocks by rank. **In parallel**, we implement a **pairs trading** strategy using K-means clustering to identify candidate pairs. Pairs are selected based on a **cointegration** interval and ranked by strength of cointegration. The top 25 pairs are then selected and traded only when their price divergence reaches the threshold determined by an Ordinary Least Squares (OLS) model trained on their historical prices.

Top 10 holdings on Average Over Testing Period:

1. gvkey: 187812, months held: 48,
2. gvkey: 22927, months held: 45,
3. gvkey: 150481, months held: 40,
4. gvkey: 20088, months held: 40,
5. gvkey: 147849, months held: 37,
6. gvkey: 9619, months held: 36,
7. gvkey: 18430, months held: 36,
8. gvkey: 166562, months held: 36,
9. gvkey: 260778, months held: 36,
10. gvkey: 118264, months held: 36,



Data & Methodology

Data Scope

- North American market (**U.S. equities**)
- Quantitative signals + text-based sentiment (SEC 10-K / 10-Q filings)

Model Selection Criteria

- **Efficiency:** Fast training & scalable
- **Diversity:** Multiple output signals & strategies
- **Compatibility:** Structured + unstructured data integration

Machine Learning Framework

- **XGBoost:** Predicts next-month returns; handles large, high-dimensional features efficiently
- **Hidden Markov Model (HMM):** Detects hidden market regimes for dynamic strategy adaptation
- **FinBERT Transformer:** Extracts sentiment from SEC earning reports using domain-specific language modeling (For US companies only)
- **K-Means Clustering (for pairs trading):** Identifies stock clusters and cointegration pairs for relative-value trades. To group stocks into clusters, the features selected as inputs to the K-Means Clustering model were selected with the goal to combine **risk-based metrics** (e.g., beta, idiosyncratic volatility) with **fundamental and size indicators** (e.g., market equity, book-to-market, profitability) to allow K-Means to reveal economically meaningful stock segments rather than purely statistical groupings.

Training structure:

- Each child model is trained on its own, generating 10 models (one per year for 2015 to 2024).
- The HMM, FinBERT Transformer and K-Means Clustering models used an expanding window approach whereas both XGBOOSTs proved to work better by training it on the data from the previous year since going too far back in the past using the expanding window made the decision trees too reliant on older data and less adaptive to recent market changes.

Stats:

- $R^2 = 0.049$, Adjusted $R^2 = 0.041$ (for the OLS regression of Parent model)
- OOS R^2 for overall sample of ML algorithms: 0.00

Results & Performance

Performance statistics for OOS testing period = Portfolio vs S&P 500

Average annualized portfolio returns = 0.2634 vs 0.1107

Annualized portfolio standard deviation = 0.5840 vs 0.1528

Annualized Alpha (Portfolio only) = 0.3830

Sharpe Ratio = 0.5619 vs 0.7665

Information Ratio (Portfolio only) = 0.6536

Maximum drawdown = 0.5154 vs 0.2846

Maximum one-month loss = -0.1845 vs -0.1250

Portfolio Turnover (Portfolio only) = 0.5160

Discussion & Conclusion

Key Takeaways & Success:

Predictive Power Validated: The integrated multi-modal ML architecture (HMM + Quant + FinBERT) successfully identified cross-sectional and textual signals, resulting in a high Annualized Alpha of 38.30%.

Complementary Design: The strategy's design to combine a directional Long-Short core with a non-directional Pairs Trading overlay proved effective at capturing multiple dimensions of alpha

Potential Improvements

- Leverage data complementarity through a parent model capturing inter-stock dependencies.
- Incorporate a quantitative measure of portfolio risk.
- Investigate unsupervised learning and neural network approaches.

Top 10 Most Profitable Stocks (by cumulative actual return in portfolio):

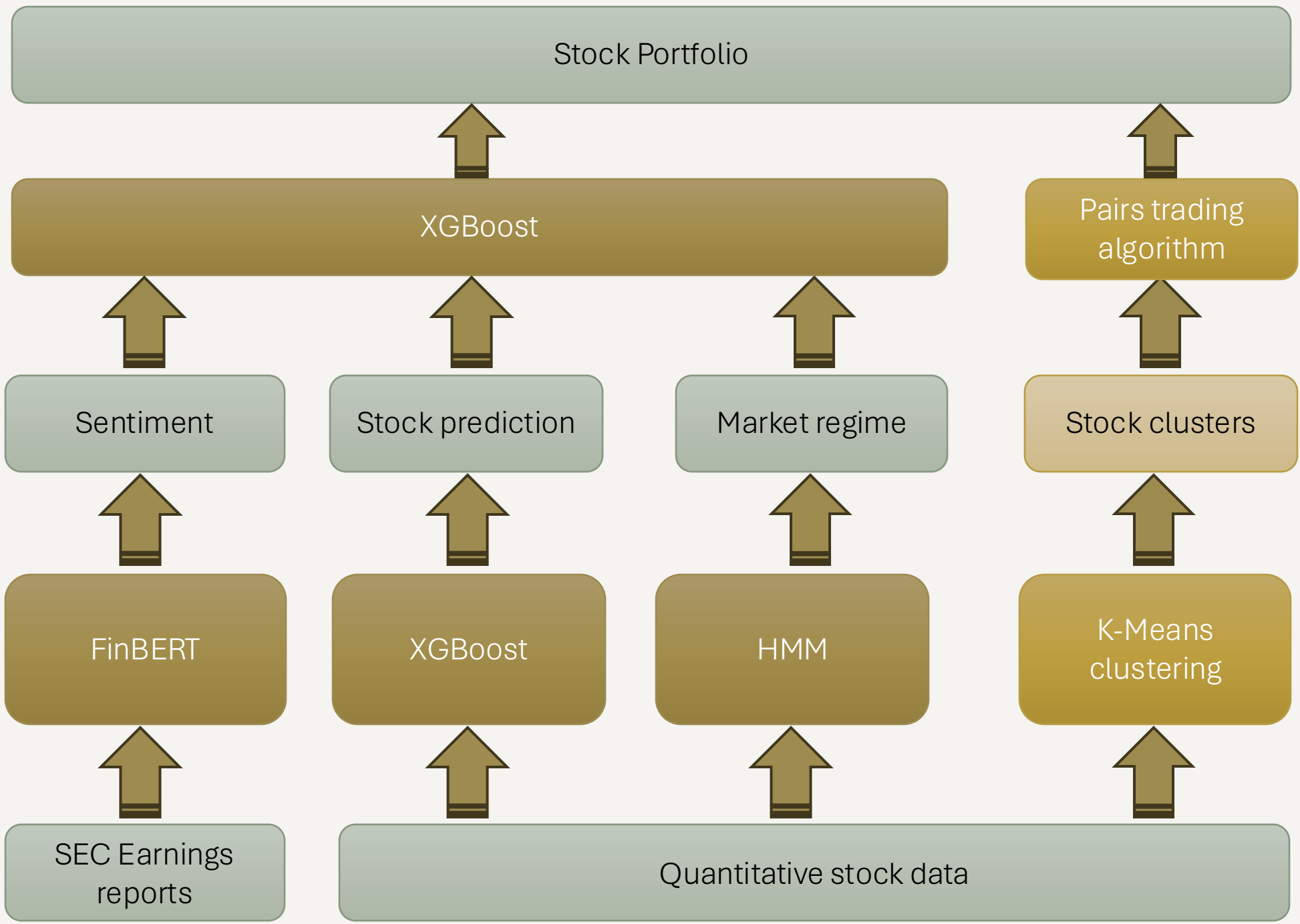
1. gvkey: 105873, cumulative contribution: 12.2029,
2. gvkey: 112759, cumulative contribution: 9.9345,
3. gvkey: 27983, cumulative contribution: 6.9919,
4. gvkey: 12711, cumulative contribution: 6.5214,
5. gvkey: 22927, cumulative contribution: 6.2670,
6. gvkey: 33114, cumulative contribution: 5.6769,
7. gvkey: 144498, cumulative contribution: 5.6342,
8. gvkey: 165563, cumulative contribution: 5.6095,
9. gvkey: 160375, cumulative contribution: 4.7424,
10. gvkey: 24966, cumulative contribution: 4.7366

Main Fundamental Signals Contributing to Portfolio Performance:

predicted_return: 3132.5266,
bearish: 554.6222,
neutral: 510.8601,
bullish: 372.9544,
hmm_state: 17.8560

Appendix A

Visualization of the architecture of the portfolio's strategy



Appendix B

Signal Complementarity

Our architecture's robustness is derived from intentionally integrating models that capture **non-redundant risk and alpha factors** across three distinct data types. This approach effectively uses all available information while simultaneously acting as a strong defense against **overfitting**.

HMM (Regime Classification)

Systematic (Macro) Risk: Market state (e.g., high volatility, mean-reverting vs. trending).

Prevents the model from applying a single set of factor weights across all market environments.

XGBoost (147 Factors)

Cross-Sectional Alpha: Factor-based anomalies (value, momentum, carry).

Prevents over-reliance on a few dominant (and potentially unstable) quantitative factors

FinBERT (Text Sentiment)

Idiosyncratic Alpha: Soft, forward-looking managerial insights and sentiment. Offers an independent, non-numerical predictive layer that acts as a check against models relying solely on historical price or balance sheet data.

Appendix C

HMM + Pairs Trading for Risk Fine-Tuning

Traditional ML Long-Short (Independent View)

Assumption: Stocks are selected based on independent expected returns. The portfolio is a collection of individually optimal positions.

Result: Primarily manages **idiosyncratic risk** (stock-specific). Ignores co-movement, leading to concentrated **correlation risk** and potential concentration of systematic exposure.

HMM + Pairs Trading (Dependent View)

Assumption: Portfolio securities are a **dependent set** where inter-stock relationships drive overall risk and return.

Result: Directly models and manages **correlation and basis risk**. Allows for **fine-tuning of portfolio risk** in the strategy's market-neutral component

- **Pairs Trading Captures Dependency:** The Pairs Trading overlay specifically targets co-moving assets (via K-means clustering and cointegration). By simultaneously going Long one leg and Short the other, this component is designed to be **market-neutral** and profit from the temporary divergence of a **dependent set** of assets. This acts as a structural hedge against broad market movements.
- **HMM Enables Dynamic Risk Fine-Tuning:** The HMM's output (Market Regime) allows the trading algorithm to **dynamically adjust parameters** for the Pairs Trading and the directional Long-Short book. For instance:
 - In Volatile Regimes** (HMM-detected), the model can tighten the cointegration window or widen the divergence threshold for pairs, reducing risk of false entry. **In Trending Regimes**, the directional Long-Short book can be given higher conviction while the pairs overlay may be scaled back.

Appendix D

Deeper look at model selection

The choice of each model in the architecture aligns the best machine learning tool with the specific financial challenge it solves.

1. Hidden Markov Model (HMM) - Regime Context

- HMMs are powerful for modeling sequences of observable events (e.g., daily returns) generated by an underlying, unobservable process—the **Market Regime**.
- **Value to Portfolio:** Unlike fixed-parameter models, HMM provides a **contextual layer** to the Meta-Model. It tells the stock-picker *what* kind of market it is operating in (e.g., a "high-volatility, low-return" state). This crucial conditioning prevents the Meta-Model from making aggressive directional bets when the HMM detects a historically challenging regime.

2. XGBoost - High-Dimensional Prediction (Meta-Model)

- As a Gradient Boosting Decision Tree, XGBoost is known for its computational efficiency and, crucially, its ability to model complex **non-linear and interacting relationships** within large datasets.
- **Value to Portfolio:** It is the ideal workhorse for our purposes:
 - **Feature Integration:** It excels at combining the high-dimensional, diverse output signals into a single, comprehensive prediction.
 - **Robustness:** Its tree-based nature makes it less susceptible to noisy data or outliers compared to linear models.

3. FinBERT - Capturing Textual Alpha

- FinBERT is a **Transformer-based Language Model** pre-trained specifically on financial text (e.g., financial news, reports). This domain-specific pre-training allows it to capture the subtle **nuances and contextual meaning** of corporate language.
- **Value to Portfolio:** It provides a unique, underutilized source of alpha: **Text as Data**. By analyzing sections like the MD&A and Risk Factors in SEC filings, FinBERT extracts a **forward-looking sentiment signal** based on soft managerial insights.

Appendix E

OOS Performance Analysis

Return Profile

- Annualized return: 26.3% vs. 11.1% (S&P 500) → strong **absolute outperformance**.
- Alpha: +38.3% → significant **excess return** beyond benchmark expectations.

Risk Characteristics

- Volatility: 58.4% vs. 15.3% → **high risk exposure**, driving performance through leverage or concentration.
- Sharpe ratio: 0.56 vs. 0.77 → **lower risk-adjusted efficiency** despite higher returns.
- Information ratio: 0.65 → **moderate consistency** in active return generation.

Drawdowns & Downside Risk

- Max drawdown: 51.5% vs. 28.5%.
- Max 1-month loss: -18.5% vs. -12.5% → **Limited downside protection** and high tail risk.

Implementation Considerations

- Turnover: 51.6% → **moderate trading intensity**, potentially eroding alpha through costs.

Overall Assessment

Strong absolute and active performance, but **high volatility and drawdowns** result in **weak risk-adjusted returns**.

Future work should focus on **risk control, diversification, and portfolio optimization**.